Modeling electrical losses in CsPbI3 solar cells: a guardrailed framework toward realistic performance projection (24.95% → 20.82%)

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Abstract: *Simulation records for CsPbI3 solar cells now exceed the certified experimental frontier (22.02%), but a survey of more than 250 SCAPS-1D architectures finds a quarter of MAPbI3 claims exceeding the best experiments by over 20%, with some breaching the Shockley–Queisser limit, driven by unrealistic inputs—above all the radiative-recombination default, unphysical defect floors, and isolated optimisation of interdependent parameters [1]. The optical-loss half of that diagnosis now has a realistic-projection framework [2]; this study supplies the electrical-loss half on one fully-worked stack. We screen an ITO/TiO2/CsPbI3/CBTS/Ni cell in SCAPS-1D over a guardrailed box—absorber 0.4–2.2 μm, N_A 10^{15}–3×10^{16} cm^{-3}, N_t 10^{12}–10^{15} cm^{-3}, E_g 1.65–1.75 eV with linked affinity, interfaces at or above a 10^{10} cm^{-2} floor ($S \geq 100$ cm/s)—and price every assumption with computed sweeps, including two the literature leaves unpriced: a radiative-recombination coefficient of 10^{-10}/10^{-9} cm^3/s costs 3.09/5.12 points at the champion but only 0.14/1.02 at routine defects, and DFT-grade mobilities cost nothing measurable at the champion (0.00–0.01 points) and 0.32 at routine defects. The best evaluated guardrailed point reaches 24.95% (Voc 1.321 V, Jsc 21.20 mA/cm^2, FF 89.11%), but it sits at the box edge in thickness and doping and on permissive affinities, so it is an upper bound rather than a forecast; the number we would defend is the routine-defect case,* ***20.82%*** *(1.0 μm, $N_t = 10^{15}$ cm^{-3}), which drops to 20.42% at $R_s = 1$ Ωcm^2. Two screens keep the optimism in check: a detailed-balance Voc computed from the model's own absorption (1.371 V at 1.65 eV) clears the champion by 0.05 V but leaves the no-interface band with no margin anywhere, and a paired Morris analysis favours interface quality over bulk defects by 2.03 points (95% CI [0.70, 3.51]) at the defect floor while the same dose–response goes nearly flat at routine defects—so the ranking is regime-dependent, not universal. Every load-bearing choice maps to one item of the standard simulation protocol proposed in [1] (Table 18). All definition files, sweep receipts, and raw curves are archived in the public repository for independent reproduction.*

Keywords: CsPbI3; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

1.0 Introduction

All-inorganic CsPbI3 perovskite solar cells combine the thermal stability lacking in hybrid organic–inorganic perovskites with a bandgap near 1.7 eV that suits both single-junction and tandem top-cell applications [3, 4]. After years below 19%, certified laboratory efficiencies have recently passed 22%: 22.02% certified (22.60% laboratory) with a moisture-responsive propyltriethoxysilane crystallisation strategy [3], 22.05% by fluorination-assisted interfacial dipole engineering [4], 21.71% with dual-interface and bulk co-passivation at an open-circuit voltage of 1.27 V [5], and 21.43% certified for inverted architectures [6]. Each record was won at the interfaces and in the bulk defect chemistry, which motivates a simulation study that treats those exact levers honestly.

The one-dimensional solar cell capacitance simulator (SCAPS-1D) is the workhorse for such pre-experimental screening, and the ITO/TiO2/CsPbI3/CBTS (copper barium thiostannate) stack of Hossain et al. [7] is the most-prescribed CsPbI3 baseline, progressing from 17.90% over 96 screened

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transport-layer combinations [8] to 19.06% after optimisation. Subsequent simulation studies report up to 24.17% on the same stack [9] and 24.24% on ZnO/PTAA [10], while large machine-learning-coupled screens now exceed 27% on other absorbers [11, 12]. Three problems accompany this inflation. First, nearly every study optimises one variable at a time (OVAT), so parameter interactions—notably absorber thickness versus defect-limited diffusion length—are invisible by construction. Second, interface recombination is removed in at least one case (stated explicitly in [10]) and defect densities are hard to verify without published parameters, which neither of the two 24%-class studies prints; we note this as unverifiable rather than adjudicated, and say so instead of inferring intent—we report our own data the same critical way, and where we could not reproduce others for lack of published parameters, we say so instead of inferring inflation. Third, scripted sweeps mix unit systems: SCAPS definition files store SI units while script 'set' commands expect practical units, so SI-valued inputs silently inject a $10^6\times$ overdose and fabricate ghost cells (scripting conventions are documented in Appendix A).

The diagnosis of this inflation now exists. A survey of more than 250 SCAPS-1D perovskite architectures finds a quarter of MAPbI3 reports exceeding the best experiments by over 20% (over nine-tenths for MASnI3), with a quarter of reported open-circuit voltages breaching the Shockley–Queisser bound outright; the causes are unrealistic inputs—above all the zero radiative-recombination default, unphysical defect floors and mobilities, and the isolated optimisation of interdependent parameters—compounded by SCAPS's optical defaults [1]. The same group has since supplied the optical-loss half of the fix, a wavelength-dependent framework toward realistic projection [2], together with a five-item standard simulation protocol: realistic inputs including the radiative coefficient, full reporting of inputs, transfer-matrix effective spectra, wavelength-dependent absorption, and an SQ check with explained deviations.

What is missing is the electrical-loss half, worked end-to-end on one stack with every assumption priced and every file archived. This paper is that companion: each protocol item maps to one row of Table 18, and each overestimation cause maps to one computed sweep.

This work contributes (i) a five-factor guardrailed grid screen (96 runs) plus refinement (33 runs) locating the surveyed-affinity best at 23.50%; (ii) a joint ETL/HTL-affinity map (36 runs) locating the best evaluated guardrailed point at **24.95%** ($\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$ eV), with an exploratory 104-configuration search peaking 1.87 points lower among guardrail-satisfying runs; (iii) an interface dose–response reported in portable τ/S alongside raw densities; (iv) scripted series- and shunt-resistance accounting including the routine-defect case; (v) pricing of the radiative-recombination default and the mobility values; and (vi) a coupled E_g–χ treatment demonstrating the protocol's ban on isolated optimisation, plus a Morris screen with paired-trajectory bootstrap reported with explicit range-dependence. Run counts and receipts are itemised in Table 19; every definition file, sweep receipt, and raw curve is archived for independent reproduction.

2.0 Literature: a decade of CsPbI3 progress

Black-phase CsPbI3 has been stabilised stepwise since 2016: quantum-dot-induced α-phase retention at 10.8% [13], co-evaporated planar cells at 17.0% [14], thermodynamically stable γ-films [15], graded-gap 14.4% devices [16], and compositionally engineered 13–14% cells [17, 18], all on TiO2 with Spiro-OMeTAD, PTAA or carbon contacts [19–21]. The 2019 aziridinium result at 18.40% [22] stood for years until the 2025–2026 interface-engineering wave (Figure 1): fluorinated dipoles (22.05% [4]), dual-interface plus bulk co-passivation with Voc 1.27 V (21.71% [5]), moisture-responsive crystallisation (22.60% laboratory, 22.02% certified [3]), and stitched grain-boundary grooves in inverted cells (21.43% certified [6]). The certified frontier is therefore 22.02% (Dai; laboratory best 22.60%; Qiu's 22.05% is uncertified), and every recent jump came from defect and interface control—the precise levers a simulation study must treat honestly.

Simulation studies trace a parallel inflation curve (Table 1). The CBTS screening programme reached 17.90% over 96 transport-layer combinations [8] and 19.06% after full optimisation [7],

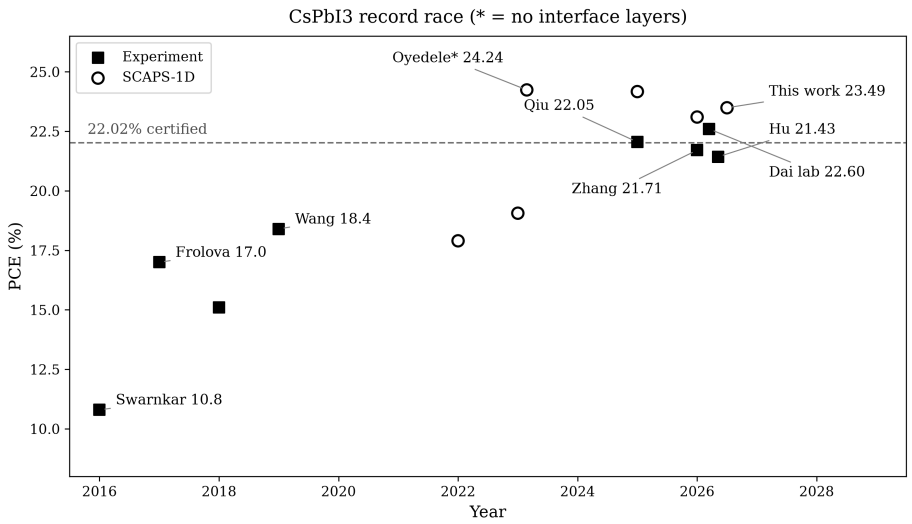


Figure 1: CsPbI3 record race. Asterisks mark simulation studies without interface layers.

Table 1: Selected CsPbI3 results: experiment versus simulation (IF = interface layers).

Study	Stack	Type	PCE (%)	IF
Swarnkar 2016	TiO2/Spiro/MoOx	Exp.	10.8	–
Frolova 2017	TiO2/PTAA	Exp.	17.0	–
Wang 2019	TiO2/Spiro	Exp.	18.4	–
Qiu 2025	DFAz dipole	Exp.	22.05	passivated
Dai 2026	TiO2/PTABr/Spiro	Exp.	22.60/22.02	passivated
Hu 2026 (Joule)	inverted HPDA	Exp.	21.43 cert.	2D-stitched
Hossain 2022	TiO2/CBTS	Sim.	17.90	on
Hossain 2023	TiO2/CBTS/Ni	Sim.	19.06	on
Ristiansyah 2026	ZnO/Spiro	Sim.	23.10	on
Nazli 2025	TiO2/CBTS/Ni	Sim.	24.17	unclear
Oyedele 2023	ZnO/PTAA	Sim.	24.24	OFF
This work	TiO2/CBTS/Ni	Sim.	24.95	on

the baseline we reproduce. Later claims report higher efficiencies on differing stacks: 23.10% with ZnO/Spiro in one report [23], 24.17% on our exact stack [9], and 24.24% on ZnO/PTAA without interface layers (stated) [10]. The broader SCAPS literature shows the same pattern across absorbers—MAPbI3 at 19.30% with inorganic contacts [24], gradient-doped FAPbI3 at 20.34% [25], a 28.38% bifacial triple-cation cell [12], and an explicitly idealised 40.17% MASnI3 result that exceeds the Shockley–Queisser bound [26, 27]. Section 11.0 states the common reporting standard these numbers would need as a concrete checklist—interface status with dose level, defect floors with cross-sections—and demonstrates it on our own cell, since the published parameter tables are too incomplete to re-place anyone else's.

3.0 Simulation framework and guardrails

Devices were simulated in SCAPS-1D v3.3.10 under AM1.5G illumination (100 mW/cm^2) at 300 K unless stated. The baseline reproduces Hossain et al. [7]: ITO front contact (4.0 eV), 30 nm TiO2 electron transport layer (donor density 9×10^{16} cm^{-3}), CsPbI3 absorber (800 nm initial, $E_g = 1.694$ eV, $\chi = 3.95$ eV, $N_A = 10^{15}$ cm^{-3}, $N_t = 10^{15}$ cm^{-3}), 100 nm CBTS hole transport layer ($N_A = 10^{18}$ cm^{-3}), and Ni back contact (5.5 eV). Crucially, neutral interface recombination

blocks ($N = 10^{10}$ cm^{-2}, $\sigma = 10^{-19}$ m^2) are present at both heterojunctions, and the absorber uses the coefficients-A-and-B absorption model ($A = 10^7$ m^{-1}eV$^{-1/2}$) so that bandgap sweeps act on the absorption edge rather than a frozen file spectrum. Our unmodified baseline yields 18.99% (Voc 1.122 V, Jsc 19.68 mA/cm^2, FF 86.00%), reproducing the published 17.9–19.06% band; against the printed 17.90% device (Voc 0.997 V, Jsc 21.07 mA/cm^2, FF 85.21% [8]) ours trades $\sim$1.4 mA/cm^2 of current for $\sim$0.12 V of voltage, as expected from added interface recombination with lower bulk defects—a single-scalar match is not claimed as validation. Optical settings: the definition file applies a flat 10% front reflection with no dispersive parasitic absorption in ITO or TiO2, and the 80% back-transmission filter faces an opaque Ni contact, i.e. no back reflector—both choices are conservative-or-neutral for Jsc except in the blue, where the flat QE plateau is an upper envelope (Section 6.0).

Guardrails enforced throughout: bulk defect floor 10^{12} cm^{-3} (single-crystal grade; nothing lower), absorber doping ceiling 3×10^{16} cm^{-3}, an interface floor of 10^{10} cm^{-2} that is part of the stated box rather than a post-hoc filter (dose points below it are audit, not candidates), conduction-band offset held near zero by linking χ to E_g ($\chi = 3.95 + 0.5(E_g - 1.694)$), and two energetic screens applied on inspection: a run with Voc above the detailed-balance value computed from the model's own Eg-sqrt absorption (1.371 V at 1.65 eV with $A = 10^7$ m^{-1}eV$^{-1/2}$ and 2.2 μm single-pass absorptance; the model sets $k_{rad} = 0$, priced in Section 6.2) or FF above Green's ideal-diode value at the computed Voc (90.5% at the joint optimum) would be discarded. Box constraints above are enforced per run by construction; these two energetic screens are *not* wired into the batch scripts—no run in the archive was rejected by them—so they are an audit applied after the fact, and we say so rather than claim an automated gate. The $E_g/q - 0.28$ V rule of earlier drafts is retired: it sat above the radiative limit and could not adjudicate anything. Series and shunt resistances are scriptable in SCAPS 3.3.10 (`set external.Rs/Rsh`; `scriptdescription3310.txt`), so this section reports computed parasitic sweeps. Sweeps ran in parallel on four workers with checkpointed receipts; of 132 grid runs, 126 converged and enter the analysis (6 runs excluded and listed in the receipts: three Round-A runs—A010 (1.0 μm, $N_A = 10^{15}$ cm^{-3}, $N_t = 10^{14}$, $E_g = 1.694$ eV), A081 (2.0 μm, $N_A = 10^{15}$ cm^{-3}, $N_t = 10^{14}$) and A051 (1.8 μm, $N_A = 10^{15}$ cm^{-3}, $N_t = 10^{12}$)—and three Round-B runs, all at 1.8 μm thickness. All six are missing-result-file worker crashes, not solver divergence, and none is a top-ten entry; because they cluster at low N_A we cannot rule out that region being harder to converge, and we note it rather than claim the exclusions are unbiased. A duplicate champion run reproduces the coarse-grid value exactly, confirming determinism; a confirmatory run at finer IV step gives 23.4874% at the surveyed affinity (FF 89.78%) against its coarse partner 23.5041%, a 0.017-point IV-step uncertainty against which all reported differences should be read. SCAPS batch scripts expose no mesh control, so no mesh-refinement study is possible in this harness; results are quoted to two decimals in the text; four decimals appear only where a receipt identity or a matched-pair difference below 0.05 points is being shown. Absorber densities of states are capped at $N_c = 1.1 \times 10^{19}/N_v = 8.2 \times 10^{19}$ cm^{-3}, ten times below the source paper's printed parameter table (10^{20} cm^{-3}, above solid density and hence unphysical).

SCAPS-1D solves the Poisson equation coupled to electron and hole continuity with drift-diffusion transport [28, 29]:

$$-\frac{\partial}{\partial x}\left(\varepsilon(x)\frac{\partial V}{\partial x}\right) = q[p - n + N_D^+ - N_A^- + p_t - n_t], \tag{1}$$

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G_n - R_n, \quad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G_p - R_p, \tag{2}$$

with Shockley–Read–Hall recombination through the specified bulk and interface defect levels. Optical generation uses the Eg-sqrt (Tauc) model $\alpha = A\sqrt{h\nu - E_g}$, so bandgap sweeps move the absorption edge self-consistently. Full input parameters are listed in Tables 2 and 3, with every guardrail and its justification in Table 4; transport-layer screenings that motivate the TiO2/CBTS choice are reviewed in [8, 30].

Table 2: Layer input parameters (script units; *unmodified baseline* values as reproduced from Hossain et al. [7]: $\chi_{\mathrm{TiO2}} = 4.0$ eV, $N_{D,\mathrm{ETL}} = 9\times10^{16}$ cm^{-3}. Screening moves to $N_D = 9\times10^{17}$ and adopts $\chi_{\mathrm{TiO2}} = 3.8$ eV; those values appear in Table 5, not here). Mobilities in cm^2/Vs; $v_{th} = 10^5$ m/s all layers; contacts $S_n = S_p = 10^5$ m/s with Ni 5.5 eV / ITO 4.0 eV; front reflection filter 10%, back transmission filter 80%; radiative (K_{rad}) off by default and priced in Section 6.2; Auger recombination off.

Layer	d	E_g	χ	ε_r	Doping	N_t	μ_n/μ_p	N_c/N_v; defect σ_n/σ_p, E_t
CBTS HTL	100 nm	1.9	3.6	5.4	N_A 1e18	1e15	30/30	2.2e18/1.8e19; 1e-20/1e-19, 1.5 eV
CsPbI3	800 nm	1.694	3.95	6.0	N_A 1e15	1e15	25/25	1.1e19/8.2e19; 1e-19/1e-19, 0.6 eV
TiO2 ETL	30 nm	3.2	4.0	9.0	N_D 9e16	1e15	20/20	2.0e18/1.8e19; 1e-19/1e-18, 1.6 eV

Table 3: Interface defect layers (both heterojunctions): neutral single level, $\sigma = 10^{-19}$ m^2 all sides, $E_t = 0.6$ eV.

Interface	N (cm^{-2})	Sweep range (this work)
CBTS/CsPbI3	1e10	10^8–10^{12} (+ ETL/HTL-only, $\sigma\times10$)
CsPbI3/TiO2	1e10	10^8–10^{12} (+ ETL/HTL-only, $\sigma\times10$)

Table 4: Guardrail values and what anchors them. "Assumption" means a modelling choice with no direct measurement behind it.

Guardrail	Value	Anchor
Bulk defect floor	$N_t = 10^{12}$ cm^{-3}	assumption (single-crystal grade)
Interface density	$N_{if} = 10^{10}$ cm^{-2}	assumption; $S \approx 100$ cm/s (demanding passivation literature)
Capture cross-section	$\sigma = 10^{-19}$ m^2	assumption (SCAPS-typical; swept $\times0.1$–$\times10$)
Trap level	$E_t = 0.6$ eV mid-gap	assumption (0.3–0.9 eV sweep: $\leq$0.012 pts)
V_{oc} ceiling	1.371 V at $E_g = 1.65$ eV (1.418 V at 1.70)	detailed-balance, model absorption ($k_{rad} = 0$)
Radiative coefficient	$k_{rad} = 0$ (priced: $-3.09/-5.12$ pts at $C_r = 10^{-10}/10^{-9}$; $-0.14/-1.02$ routine)	assumption, priced Sec. 6.2; Eq. 5 $\approx$ omission
Mobilities	25/25 absorber (priced: $\leq$0.01 champion, -0.32 routine at 10/5)	assumption, priced Sec. 6.2
FF ceiling	Green's ideal at computed V_{oc}	analytic bound
R_s, R_{sh} basis	scripted sweeps, reported	computed, not assumed zero

3.1 Protocol compliance

This study is designed as a worked implementation of the standard simulation protocol proposed in [1]; Table 18 carries the item-by-item mapping, summarised here. (i) Realistic inputs: every layer, defect, and contact parameter is printed with its source and operating-point validity in Tables 2–4, with the density-of-states cap (ten times below the source paper's printed, above-solid-density values) and the fabrication-anchored doping ceiling stated rather than buried. (ii) Full reporting: bulk and interface defects are reported as N *with* σ alongside the portable τ/S equivalents (Table 13), and the radiative coefficient is reported as a swept assumption, not a default (Section 6.2). (iii) Effective spectra and absorption: we retain the Eg-sqrt model with the edge-moving coefficient so gap sweeps act on absorption rather than a frozen spectrum, and we carry the optical uncertainty instead of hiding it—a flat 10% front reflection with no dispersive parasitics (an upper envelope in the blue, $\sim$1 mA/cm^2 at stake) plus the 1.19 mA/cm^2 QE-integral-vs-IV gap reported as Jsc uncertainty; the wavelength-dependent transfer-matrix half is cited, not re-derived [2]. (iv) SQ check: the model-derived 1.371 V radiative bound screens every run (Section 7.0), and the no-interface band's lack of margin is reported as disqualification, not rank order. (v) Explained deviations: Table 17 attributes the residual excess over measured devices to named causes (fill factor, gap) with the un-isolated systematics stated. Where the protocol asks for an explanation and we have only a bound, we say so.

4.0 Grid screen and joint-optimum champion

Round A explored the full cross-product of absorber thickness (1.0–2.0 μm), acceptor density (10^{15}–10^{16} cm^{-3}), bulk defect density (10^{12}–10^{14} cm^{-3}), ETL donor density (9×10^{16}–9×10^{17} cm^{-3}), and bandgap (1.694/1.65 eV with linked affinity)—96 runs (93 converged, 3 worker crashes) in a single parallel batch, rather than sequential OVAT slices. The winner (2.0 μm, $N_A = 10^{16}$ cm^{-3}, $N_t = 10^{12}$ cm^{-3}, $N_{D,ETL} = 9\times10^{17}$ cm^{-3}, $E_g = 1.65$ eV) reached 23.47%. Full parameters are collected in Table 5. Round B refined thickness (1.8–2.2 μm), gap (1.65–1.72 eV), and doping around this point (36 runs, 33 converged) and reached 23.50% (2.2 μm, $N_A = 3\times10^{16}$ cm^{-3}, $N_t = 10^{12}$ cm^{-3}, $E_g = 1.65$ eV, $\chi = 3.928$ eV). A subsequent ETL-affinity sweep showed the surveyed 4.0 eV is not optimal: 3.9 eV gives 23.80%, 3.8 eV gives 23.82–23.84% across two batches (23.8241 in `s7_r3` and `s11`, 23.8365 in `s8`; the 0.012 spread is inside the 0.017-point IV-step uncertainty above, and we quote 23.84% for the batch that produced the gap-step and E_t analysis, against the surveyed anchor 23.5041), while 4.1 and 4.2 eV collapse to 21.92% and 19.87%. Because the two affinities interact through the interface activation energy, we then mapped them jointly at guardrailed interfaces (36 runs; Table 6 prints every second χ_{CBTS} row plus the 3.95–4.10 eV extension): at fixed χ_{TiO2}, efficiency rises monotonically with χ_{CBTS} through 3.90 eV, and the peak extension (3.95–4.10 eV) turns over past $\sim$3.95 eV, so the optimum is interior, not edge-sitting. The best evaluated guardrailed point is **24.95%** at $\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$ eV (Voc 1.321 V, Jsc 21.20 mA/cm^2, FF 89.11%; Figure 2), verified digit-identical with explicitly set $N_{if} = 10^{10}$ cm^{-2} (24.9464 both ways, receipt `s22_jointIF.json`) and falling to 24.24% at 10^{11} cm^{-2} (24.2377, same receipt), confirming the interfaces are active at the optimum. An exploratory 104-configuration search (LHS plus pattern search over thickness, $\log N_A$, $\log N_t$, $\log N_D$, E_g, $\log N_{if}$, χ_{TiO2}) was run over a deliberately wider box than the guardrailed one—$\log N_{if}$ down to 10^8 and χ_{TiO2} from 3.8 eV—so that it cannot bracket the champion and is reported only as an exploration: 95 of 104 runs converged, 69 sit below the 10^{10} cm^{-2} interface floor and none samples $\chi_{\mathrm{TiO2}} < 3.8$ eV. Its overall peak is 23.76% (itself a sub-guardrail cell at $N_{if} = 4.6\times10^9$ cm^{-2}), and 23.08% among its 25 runs satisfying both the interface floor and the stated $N_A \le 3\times10^{16}$ cm^{-3} ceiling (26 meet the interface floor alone; the sampler overshot the doping ceiling by 2% in one run)—below every grid champion, but not a test of it. 24.95% therefore stands as the best of the 180 guardrail-satisfying configurations we evaluated on named geometries (Round-A 96, Round-B 36, affinity map 36, new-base 12; 174 converged), plus routine-defect and audit batches that probe rather than optimise—not as a proven global optimum.

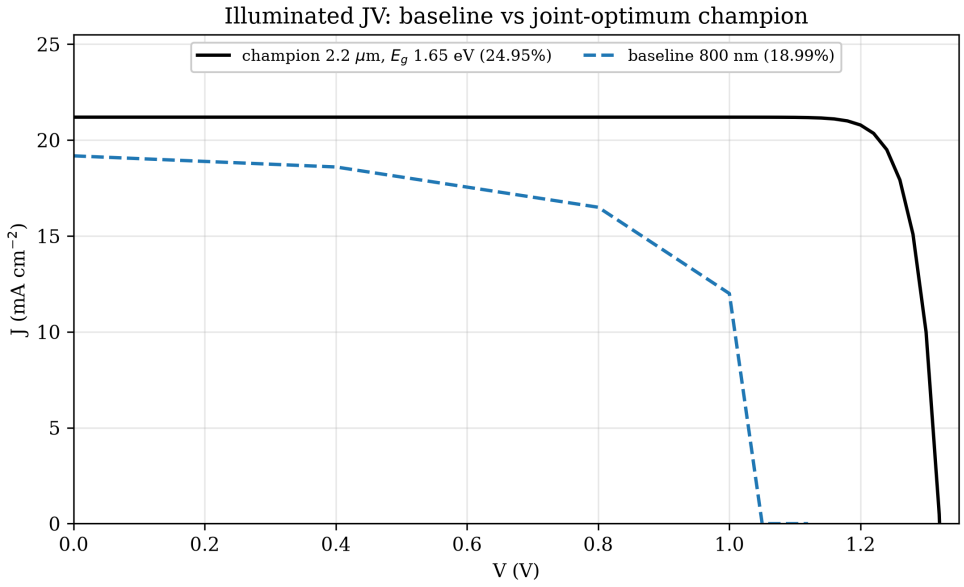


Figure 2: Illuminated current–voltage curves: baseline (800 nm, 18.99%) versus joint-optimum champion (2.2 μm, 24.95%).

Two features of this point matter. First, it sits at the corner of the allowed box in every dimension except the affinities, which are themselves aggressive ($\chi_{\mathrm{TiO2}} = 3.7$ eV below the typical 4.0–4.2 eV range; $\chi_{\mathrm{CBTS}} = 3.9$ eV above the surveyed 3.6 eV): this is a permissive-edge upper bound, and Section 9.0 prices each edge. Second, the best grid point at the literature gap $E_g = 1.694$ eV reaches 22.506%, coincidentally equal at two decimals to the 22.514% of the $E_g = 1.70$ eV attainable-gap run on a different geometry (they differ by 0.008 points, inside step uncertainty), so roughly one point comes from the gap choice alone. Every top-ten configuration carries $E_g = 1.65$ eV, gaining $\sim$1.5 mA/cm^2 over 1.694 eV; the joint optimum presents a +0.23 eV ETL-side electron spike that raises the interface recombination activation energy (Section 5.0).

From diffusion estimates ($\mu_n = 25$ cm^2/Vs, $\sigma = 10^{-19}$ m^2): $L \approx 80$ μm at 10^{12} cm^{-3}, $\approx$8 μm at 10^{14} cm^{-3}, $\approx$2.5 μm at 10^{15} cm^{-3}. The map (Figure 3) is therefore nearly thickness-insensitive at 10^{14} cm^{-3} (near-vertical contours)—a thickness collapse there cannot be diffusion-limited, so we do not claim one. The thickness optimum moves with defect level, which is why the single-variable sweep had to be repeated. A two-factor thickness $\times$ N_t factorial at the joint-optimum base gives, at $N_t = 10^{15}$ cm^{-3}, 20.11% (0.5 μm), 20.76% (0.8 μm), 20.86% (1.0 μm) and 20.81% (1.5 μm), against 21.67/22.42/22.53/22.57% at 10^{14} cm^{-3}: the same 0.7–0.9 points of thickness structure at both defect levels, with the peak moving from 1.5 μm at the cleaner film to 1.0 μm at routine defects—the interaction is real but small, and we claim only that the optimum shifts with N_t, not that thickness dominates. ETL donor density confirms the literature optimum (9×10^{17} cm^{-3}) and transport-layer thicknesses are insensitive (100/30 nm retained), both consistent with [7]. The current–voltage curve (Figure 2) is square and artefact-free, with a Voc deficit of 0.33 V against E_g/q at the joint optimum (0.40 V at the prior $\chi_{\mathrm{TiO2}} = 3.8$ eV base).

The 5.96-point stacked gain decomposes along the screening path (Table 5 parameters in sequence, illustrative ordering only): thickness 800 nm to 1.5 μm adds 0.19 points (19.18%), acceptor doping to 10^{16} cm^{-3} adds 0.93 points (20.11%), bulk-defect reduction to 10^{12} cm^{-3} adds $\sim$1.6 points ($\sim$21.7%), the joint E_g–χ move to 1.65 eV with ETL donor optimisation contributes $\sim$1.8 points (to the 23.50% surveyed best), and the joint affinity optimum adds the final 1.45 points (modulo rounding). This sequential attribution is order-dependent and illustrative, not causal. Open-circuit voltage climbs 1.122 to 1.321 V while the fill factor rises 86.0 to 89.1%, the signature of SRH-lifetime recovery plus interface-activation recovery rather than band-shift artefacts.

Table 5: Champion device parameters (script units) and result: best evaluated of 180 guardrail-satisfying configurations on named geometries, not a proven global optimum.

Layer	d	E_g (eV)	χ (eV)	Doping (cm^{-3})	N_t (cm^{-3})
CBTS HTL	100 nm	1.9	3.9	NA 1e18	1e15
CsPbI3 absorber	2.2 μm	1.65	3.928	NA 3e16	1e12
TiO2 ETL	30 nm	3.2	3.7 (joint optimum; surveyed 4.0)	ND 9e17	1e15
Interfaces (both)	–	–	–	–	1e10 cm^{-2}
Result	Voc 1.321 V, Jsc 21.20 mA/cm^2, FF 89.11%, PCE 24.95%				

Table 6: Joint affinity map at guardrailed interfaces (PCE %; champion geometry; peak extension 3.95–4.10 eV turns over past ∼3.95 eV).

$\chi_{\mathrm{CBTS}} \setminus \chi_{\mathrm{TiO2}}$	3.7	3.8	3.9	4.0
3.50	22.10	22.10	22.10	22.07
3.60	23.83	23.82	23.80	23.50
3.70	24.67	24.64	24.53	23.77
3.80	24.89	24.86	24.67	23.81
3.90	**24.95**	24.91	24.74	23.81
3.95	24.94	24.90	–	–
4.00	24.81	24.79	–	–
4.10	23.25	23.25	–	–

Round A's top ten (Table 7) all carry $E_g = 1.65$ eV; the top seven sit at $N_t = 10^{12}$ cm^{-3} while rows 8–10 at $N_t = 10^{13}$ cm^{-3} trail by ∼0.4 points through Voc loss (1.233 to 1.221 V). Round B refined thickness (1.8–2.2 μm), gap (1.65–1.72 eV) and doping (36 runs) and moved the best from 23.47% to 23.50% (coarse grid), i.e. the landscape is converged rather than climbing.

Round B's refinement is documented in full because its plateau is itself evidence: the best ten span only 23.32–23.50% (Table 8), with thickness 1.8–2.2 μm and N_A (1–3)$\times 10^{16}$ cm^{-3} numerically indistinguishable at grid resolution once $N_t = 10^{12}$ cm^{-3} and $E_g = 1.65$ eV are fixed (the ten span 23.32–23.50%, a 0.19-point range against a 0.017-point coarse/fine IV-step uncertainty, so the leading cluster cannot be separated even though the 10th entry trails observably). The $N_t = 3 \times 10^{12}$ cm^{-3} entries trail the matched 10^{12} cells by 0.05–0.16 points (0.05 at the champion corner, 0.16 on the $N_A = 10^{16}$ branch), confirming the defect floor binds the optimum.

Table 7: Round A leaderboard (best ten of 96; $N_{D,\mathrm{ETL}} = 9 \times 10^{17}$ cm^{-3}; $\chi_{\mathrm{TiO2}} = 4.0$ eV surveyed throughout).

d (μm)	N_A	N_t	E_g	Voc	Jsc	PCE
2.0	1e16	1e12	1.65	1.233	21.18	23.47
1.8	1e16	1e12	1.65	1.233	21.16	23.45
1.5	1e16	1e12	1.65	1.234	21.11	23.40
1.5	1e15	1e12	1.65	1.233	21.11	23.29
2.0	1e15	1e12	1.65	1.233	21.18	23.29
1.0	1e16	1e12	1.65	1.234	20.90	23.18
1.0	1e15	1e12	1.65	1.233	20.90	23.11
1.8	1e16	1e13	1.65	1.221	21.16	23.04
2.0	1e16	1e13	1.65	1.219	21.18	23.03
1.5	1e16	1e13	1.65	1.223	21.11	23.03

Table 8: Round B leaderboard (best ten of 36; $N_{D,\mathrm{ETL}} = 9 \times 10^{17}$ cm^{-3}).

d (μm)	N_A	N_t	E_g	PCE
2.2	3e16	1e12	1.65	23.50
2.0	3e16	1e12	1.65	23.49
2.2	1e16	1e12	1.65	23.48
2.0	1e16	1e12	1.65	23.47
1.8	3e16	1e12	1.65	23.47
2.2	3e16	3e12	1.65	23.45
1.8	1e16	1e12	1.65	23.45
2.0	3e16	3e12	1.65	23.44
1.8	3e16	3e12	1.65	23.42
2.2	1e16	3e12	1.65	23.32

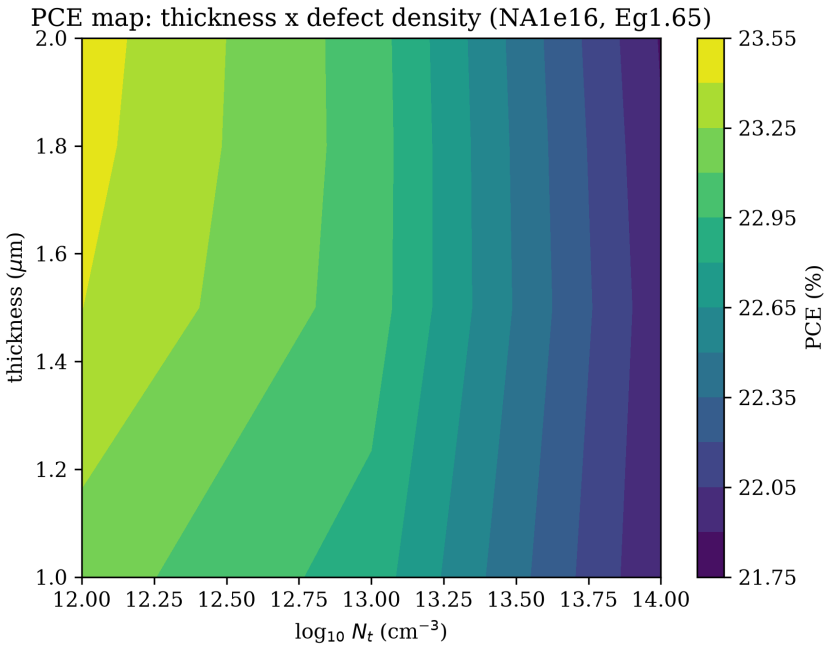


Figure 3: Efficiency map over absorber thickness and bulk defect density (slice at $N_A = 10^{16}$ cm^{-3}, $E_g = 1.65$ eV).

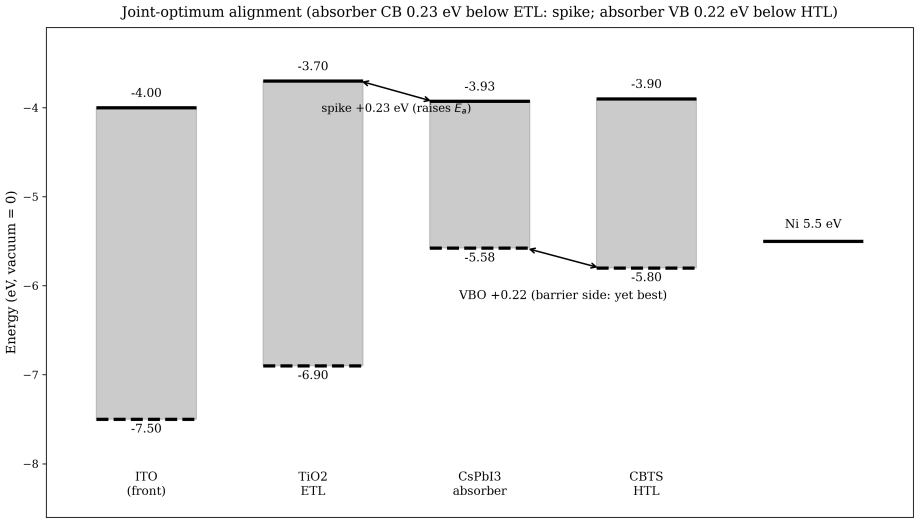


Figure 4: Equilibrium band alignment of the champion stack from input parameters (energies in eV vs vacuum).

5.0 Band alignment

Figure 4 constructs the equilibrium alignment from the joint-optimum parameters. The absorber conduction edge (-3.928 eV) lies 0.23 eV *below* the TiO2 edge (-3.70 eV), so electrons face a spike entering the ETL. The data support reading this as recombination physics, not transport: the interface recombination activation energy $E_a = E_{c,\mathrm{abs}} - E_{v,\mathrm{HTL}}$ grows from 1.47 eV at $\chi_{\mathrm{CBTS}} = 3.5$ to 1.87 eV at 3.9 eV, and efficiency rises monotonically through zero hole-side offset (22.10% to 24.95% at $\chi_{\mathrm{TiO2}} = 3.7$; Table 6)—a transport reading predicts the barrier end should be worst, and it is the best. On the hole side, the absorber valence edge (-5.578 eV) lies 0.22 eV *above* the CBTS edge (-5.80 eV) at the optimum, past the ideal -0.3 to -0.1 eV window on the barrier side—yet efficiency peaks there, which a transport reading cannot produce. The same activation logic applies to the ETL side, which is why the spike buys rather than costs efficiency. Both optimum affinities sit outside the surveyed values ($\chi_{\mathrm{TiO2}} = 3.7$ below the typical 4.0–4.2 eV; $\chi_{\mathrm{CBTS}} = 3.9$ above the surveyed 3.6 eV) and are reported as joint-optimum adoptions priced in Section 9.0, not surveyed values.

6.0 Device characterisation

The quantum efficiency (Figure 5) is flat at ∼89–90% from 300 nm with the absorption edge red-shifted from 731 nm (baseline, 1.694 eV) to 752 nm (champion, 1.65 eV), directly explaining the Jsc gain. The UV plateau assumes the definition file's flat 10% front reflection with no dispersive parasitic absorption in ITO or TiO2, so it is an upper envelope in the blue rather than a prediction; with TiO2 absorbing above 3.2 eV, of order 1 mA/cm^2 of the reported Jsc is at stake in the UV, which is large next to the points debated here and is stated as an uncertainty, not corrected. Integrating the computed QE against the AM1.5G spectrum gives 22.39 mA/cm^2 against the 21.20 mA/cm^2 IV value; the 1.19 mA/cm^2 gap is carried as Jsc uncertainty with the conservative IV value headlined. Capacitance–voltage analysis at 1 MHz (Figure 6) shows rising capacitance with forward bias (31 to 39 nF/cm^2 over 0–0.5 V). The low-bias $1/C^2$ branch is linear ($R^2 = 1.000$) with the normal sign for a p-type absorber; its slope gives $N_A = 2.93 \times 10^{16}$ cm^{-3}, essentially the 3×10^{16} cm^{-3} input, extrapolating to $V_{bi} = 1.30$ V—a solver self-consistency check on the doping from its own electrostatics, not an independent validation. The SCAPS-extracted width (0.29 to 0.22 μm) sits an order of magnitude below the 2.2 μm absorber throughout (the parallel-plate check gives 0.17 μm at 0 V, consistent

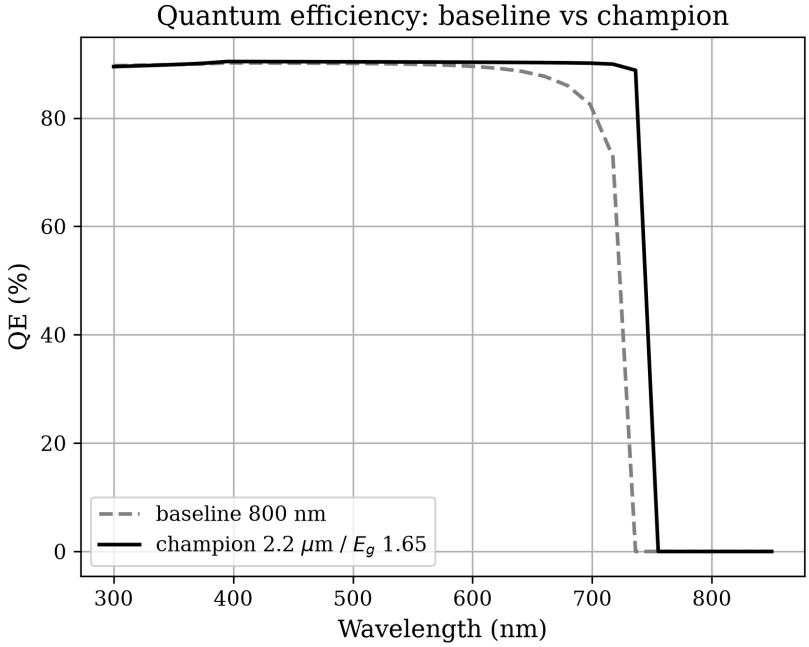


Figure 5: Quantum efficiency of baseline versus champion; the edge shifts 731 to 752 nm.

in order with extra deep-level capacitance unaccounted; this check therefore confirms the sign and linearity of the $1/C^2$ branch, not the absorber thickness). The illuminated band diagram at short circuit (Figure 7) exhibits clean quasi-Fermi splitting, and generation exceeds recombination by at least two orders of magnitude at every mesh point of the full 2.33 μm short-circuit profile (minimum ratio $\sim 3\times 10^2$ at $x \approx 0.2$ μm near the back contact, rising above 10^4 at both contacts; Figure 8). All profile statements in this section are at short circuit; nothing here constrains the maximum-power or open-circuit regime.

Temperature dependence (Figure 9, Table 9) follows the established trend [7]: efficiency falls from 25.88% at 275 K to 18.13% at 475 K, driven by Voc (1.357 to 1.038 V) and FF while Jsc stays pinned at $\sim$21.20 mA/cm^2, i.e. -0.16%/K relative (-0.039 points/K) over the 300–475 K span. The 275 K fill factor (89.96%) sits *below* Green's ideal-diode value at that Voc and temperature ($\sim$91.3%), so no artefact needs invoking; likewise the joint optimum's 89.11% is about 1.4 points below ideal (90.5%), leaving genuine FF headroom short of series resistance. The 275 K Voc (1.357 V) sits under the 275 K radiative limit (1.397 V, recomputed at that temperature), so the screen holds across the sweep. The 475 K end of the sweep ignores the temperature dependence of E_g and black-phase stability, so it is a solver stress test, not a prediction.

6.1 Parasitic-resistance accounting

Series and shunt resistances are scriptable in SCAPS 3.3.10 (`set external.Rs/Rsh`; `scriptdescription3310.txt`). Computed scripted sweeps at the joint-optimum base give 24.73% at $R_s = 0.5$ Ωcm^2, **24.52%** at $R_s = 1$ Ωcm^2, and 24.09% at $R_s = 2$ Ωcm^2 (Table 10); a lumped-series analytic derate $\eta = \eta_0/[1 + J_{sc}R_s/(10^3 FF_0 V_{oc})]$—the 10^3 putting FF_0V_{oc} on the same millivolt scale as $J_{sc}R_s$, with $FF_0 = 0.891083$ the 89.11% of Table 5 as a fraction—tracks the computed sweep to within 0.011 points at all three R_s (24.7237/24.5050/24.0790 against 24.7311/24.5160/24.0866). Shunt paths cost 0.14 points at $R_{sh} = 10^4$ Ωcm^2 at the prior base; combined $R_s = 1/R_{sh} = 10^4$ at the joint optimum gives 24.38%. The best point therefore survives parasitics, but only just: at $R_s = 2$ Ωcm^2 the margin over the 22.02% certified record is about two points. At routine defects $R_s = 1$ costs 0.40 points (20.82 to 20.42%).

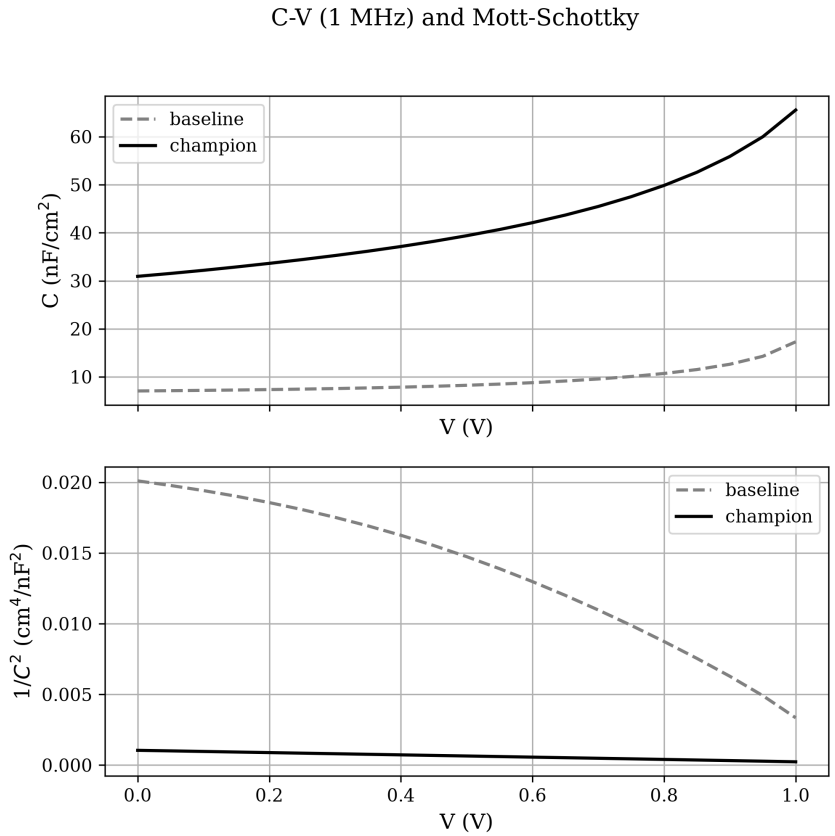


Figure 6: Capacitance–voltage (1 MHz) and Mott–Schottky plots for baseline and champion.

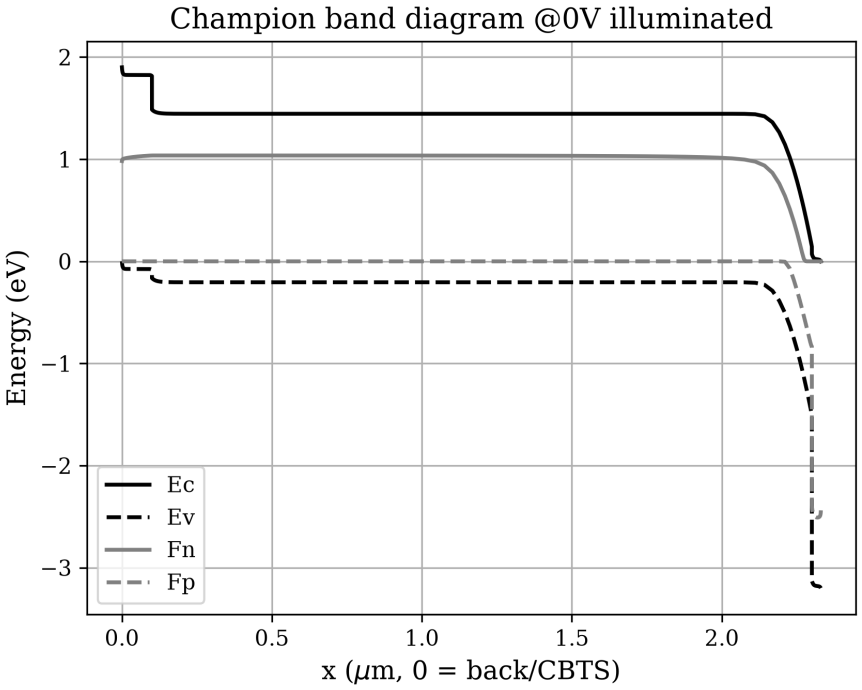


Figure 7: Illuminated band diagram of the champion at short circuit (x=0 at the back contact).

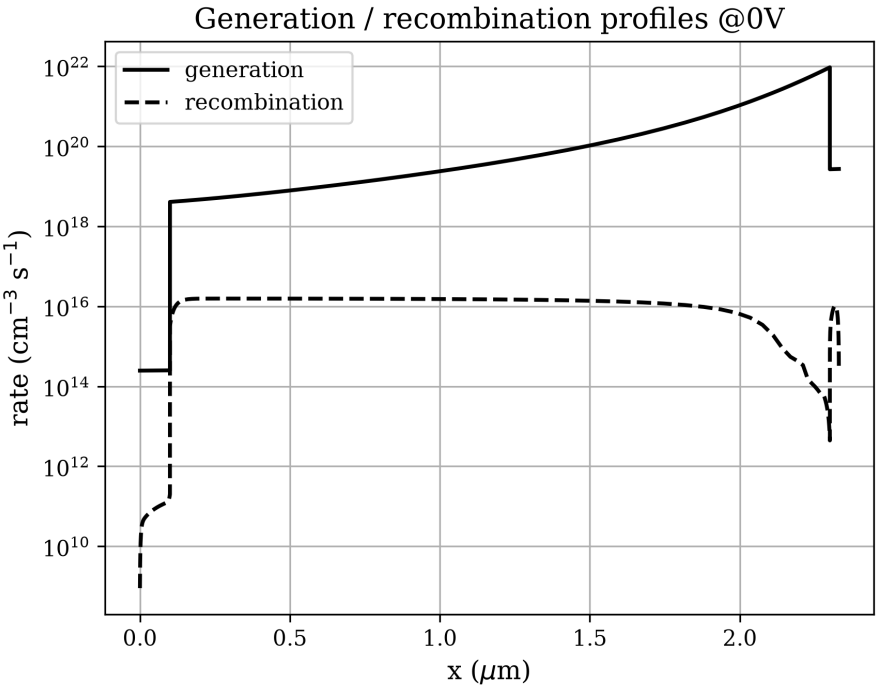


Figure 8: Generation and recombination profiles at short circuit (log scale).

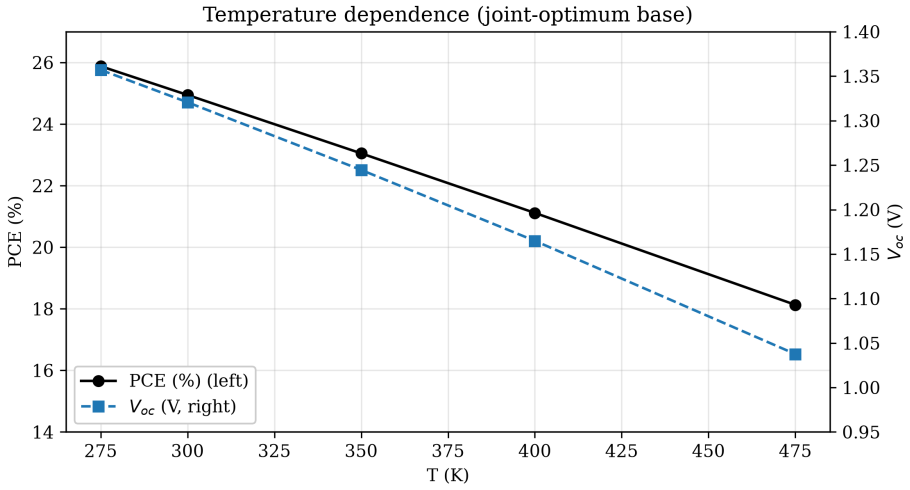


Figure 9: Efficiency and open-circuit voltage versus operating temperature at the joint-optimum base ($R_s = 0$); the 475 K end is a solver stress test that ignores $E_g(T)$ and black-phase stability.

Table 9: Temperature dependence at the joint-optimum base (1 sun; T300 = 24.9464).

T (K)	Voc (V)	Jsc (mA/cm^2)	FF (%)	PCE (%)
275	1.357	21.20	89.96	25.88
300	1.321	21.20	89.11	24.95
350	1.245	21.20	87.37	23.05
400	1.165	21.20	85.51	21.12
475	1.038	21.20	82.43	18.13

Table 10: Best point under computed scripted parasitics (joint-optimum base; all values computed, none derated).

Parasitics	PCE (%)	vs 22.02% cert.	Status
$R_s = 0$ (box maximum)	24.95	+2.93	SCAPS box maximum
$R_s = 0.5\ \Omega\text{cm}^2$	24.73	+2.71	optimistic good contact
$R_s = 1\ \Omega\text{cm}^2$	24.52	+2.50	optimistic; ITO sheet resistance is typically higher
$R_s = 2\ \Omega\text{cm}^2$	24.09	+2.07	margin narrowing
$R_s = 1/R_{sh} = 10^4$ combined	24.38	+2.36	computed, joint base

Table 11: Pricing the $k_{rad} = 0$ and mobility assumptions (computed scripted runs; Δ against the $k_{rad} = 0$ anchor of each base). The defended number is robust: 20.82% goes to 20.68/19.80% at literature C_r; SCAPS's own Eq. 5 value (3.5×10^{-14}) is equivalent to omission.

Run	Voc (V)	Jsc (mA/cm^2)	PCE (%)	Δ (pts)
Joint optimum ($k_{rad} = 0$)	1.321	21.20	24.95	–
$+C_r = 10^{-10}$ cm^3/s	1.163	21.08	21.86	−3.09
$+C_r = 10^{-9}$ cm^3/s	1.113	20.43	19.83	−5.12
μ_n/μ_p 10/5	1.321	21.20	24.94	−0.00
μ_n/μ_p 5/2	1.321	21.20	24.94	−0.01
Routine defects ($k_{rad} = 0$)	1.147	20.77	20.82	–
$+C_r = 10^{-10}$ cm^3/s	1.141	20.73	20.68	−0.14
$+C_r = 10^{-9}$ cm^3/s	1.115	20.45	19.80	−1.02
μ_n/μ_p 10/5	1.148	20.60	20.50	−0.32

6.2 Radiative-recombination and mobility pricing

The survey identifies the zero radiative-recombination default as the primary Shockley–Queisser-violation mechanism: raising C_r from 10^{-10} to 10^{-9} cm^3/s collapses their demo cell from 21.08% to 13.33% [1]. Our campaign sets $k_{rad} = 0$ throughout, so this section prices that choice with scripted `set layer2.radiative` runs at both headline bases (receipts in Table 19; `s23_krad_mu.json`). The script key was validated Appendix-A style: `radiative 0` and `mun/mup 25/25` reproduce the joint and routine receipts digit-identically (24.9464% and 20.8197%), confirming the key paths and the practical units (cm^3/s, cm^2/Vs); the 10^{-16} point reproduces the baseline to 0.0001, bracketing SCAPS's own Eq. 5 value (3.5×10^{-14} cm^3/s at our N_c/N_v, E_g, and thickness), which is therefore equivalent to omitting radiative recombination entirely.

Three findings (Table 11). First, the $k_{rad} = 0$ assumption is worth 3–5 points at the defect floor, almost entirely through Voc (1.321 to 1.163/1.113 V)—the same channel the survey flags, and the reason the champion's 0.05 V radiative margin must be read as model-conditional. Second, the cost is regime-dependent like everything else in this study: at routine defects, where SRH recombination dominates, the same C_r values cost only 0.14/1.02 points, so the defended 20.82% number is robust to the radiative assumption within about a point. Third, the mobility assumption is nearly free at the champion (0.00–0.01 points for DFT-grade 10/5 and 5/2 against our 25/25) and small at routine defects (0.32 points, via Jsc and FF): transport is not what holds this cell up, consistent with the interface-led Morris ranking. One run (R_rad1em10) needed a solo retry after a worker crash, noted in the receipts; no result in this table is a crash artefact.

7.0 Interface dose–response and the limits of auditing others

Deleting the interface blocks (audit pairs in Table 14, dose–response plotted in Figure 10) inflates efficiency by 1.26–1.66 points absolute at the surveyed base (23.40 to 24.87%, 23.47 to 24.73%, 23.50 to 25.16%), almost entirely through a 0.13 V Voc rise as interface recombination vanishes. But

the binary ON/OFF comparison understates the audit: the full dose–response at the prior adopted base (Table 12) runs 25.11% at 10^8 cm^{-2}, 24.77% at 10^9 cm^{-2}, 23.84% at the 10^{10} cm^{-2} guardrail, 22.62% at 10^{11} cm^{-2}, and 21.20% at 10^{12} cm^{-2}. Because interface density and cross-section enter only as the product $N\sigma$, Table 13 restates the guardrailed regime in the portable metrics other studies can compare against: bulk lifetime $\tau = 1/(\sigma v_{th} N_t)$ and interface velocity $S = \sigma v_{th} N_{if}$ (with $v_{th} = 10^5$ m/s). Our guardrail is $\tau = 100$ μs with $S = 100$ cm/s. The decisive screen, however, is energetic, not parametric: the detailed-balance Voc computed from the model's own Eg-sqrt absorption is 1.371 V at 1.65 eV (1.418 V at 1.70 eV), and the no-interface cells we computed (1.363–1.382 V) sit within 0.011 V of that radiative limit—the top one above it, the rest with no margin either—because with $k_{rad} = 0$ in the model there is nothing to hold the quasi-Fermi splitting below the radiative bound. A band that hugs the limit is not evidence of a working cell. The published 24.17% [9] and 24.24% [10] values are therefore not placed on our curve at all: their stacks, gaps, thicknesses and defect parameters all differ from ours, neither paper prints a complete parameter table from which τ or S could be reconstructed, and we state this as a limitation rather than a finding.

7.1 Where the missing points go

The inflation is almost entirely a Voc effect (+0.13 V at the champion cell, +0.148 V at 1.5 μm) with Jsc unchanged to within 0.02 mA/cm^2: removing the 10^{10} cm^{-2} interface states eliminates the dominant SRH channel at the heterojunctions, unpinning the quasi-Fermi splitting. The no-interface Voc values (1.363–1.382 V) all sit within 0.011 V of the 1.371 V detailed-balance limit computed from the model's own absorption, the top one above it: with $k_{rad} = 0$ there is no radiative channel to pin the splitting, so none of these cells carries margin and none is trusted, regardless of rank order. Fill factor moves non-monotonically (89.86% down to 85.25–87.05% across the three audited cells) as the recombination profile reshapes. The no-interface 87.05% at Voc 1.363 V sits 3.7 points below Green's ideal (90.8%): the archived JV curve stays flat to 1.16 V with no kink or S-shape, so there is no visible extraction barrier, but the knee remains SRH-rounded by the surviving bulk channel—removing interfaces unpins Voc without idealising the diode. This bounds the "no extraction barriers" claim to current collection, not to fill factor. Asymmetric tests show both heterojunctions matter: 10^{12} cm^{-2} on the ETL side alone gives 21.35%, on the HTL side alone 21.20%. Raising the capture cross-section 10× at fixed $N_{if} = 10^{10}$ cm^{-2} gives 22.2108%, digit-identical to the $N \times 10$ run at the same base (both 22.2108; receipts in Table 19): for the symmetric single level, rate scales exactly with the $N\sigma$ product. The same identity holds in the bulk: $\sigma \times 10$ at the N_t floor gives 23.2572% (digit-identical to $N_t \times 10$ at the same base, = 23.2572; receipts in Table 19), a 0.25-point cost against the surveyed 23.5041 anchor. (A first attempt via the script `set` key for σ neutralised the interfaces instead of scaling them; only the verified definition-file variant is reported—see Appendix A.) Bulk $N_t \times \sigma$ costs 0.25 points at the 10^{12} cm^{-3} floor but 3.12 points at routine 10^{15} cm^{-3} (20.56% at matched $\sigma \times 1$ geometry down to 17.44%), while $\sigma \times 0.1$ at the floor gains only 0.03 points (23.53%)—the floor is saturated.) The computed parasitic sweep at the adopted base is given in Table 10.

8.0 Morris sensitivity screening

Grid methods with uneven ranges cannot rank physical importance, so we ran a Morris sensitivity screen instead: $r = 8$ trajectories, $p = 4$ levels, $k = 6$ factors (absorber thickness, $\log N_A$, $\log N_t$, $\log N_{D,ETL}$, E_g with the χ link applied deterministically, and $\log N_{if}$ tied on both interfaces), 56 fresh simulations, all converged, spanning 15.6–23.3% (Table 15, Figure 11). Per decade, interface quality leads at 1.02 points (bootstrap 95% CI [0.84, 1.22]) ahead of bulk defect density at 0.68 (CI [0.35, 0.99]). Because marginal intervals overlap, we ran the sharper paired test the design allows: the per-trajectory difference $|EE_{Nif}| - |EE_{Nt}|$ averages +2.03 points with a bootstrap 95% CI of [0.70, 3.51] (6 of 8 trajectories positive)—interfaces genuinely lead *within the defect-floor regime*. Per eV,

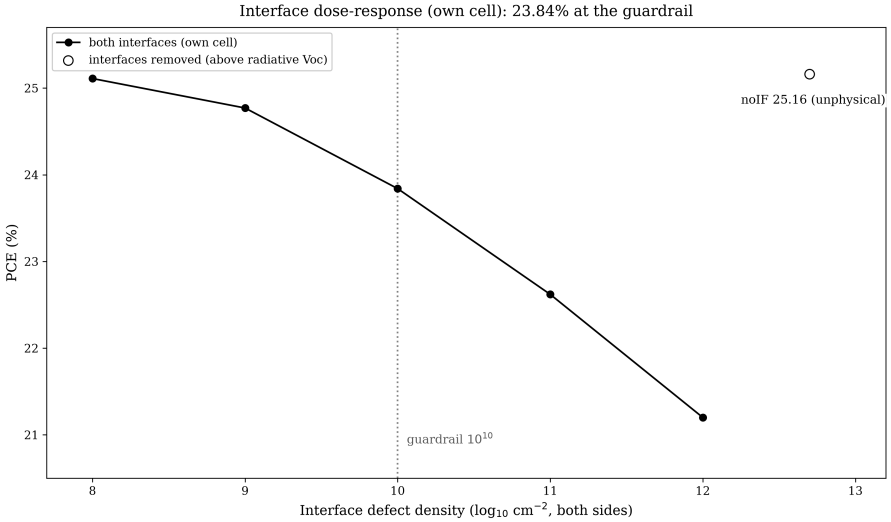


Figure 10: Same cell at graded interface quality: efficiency vs N_{if} (both sides). No literature scale bar is shown: cross-stack mapping is not claimed (Section 7.0).

Table 12: Interface dose–response (prior adopted base unless noted; both sides unless noted).

N_{if} (cm^{-2})	PCE (%)	Note
10^8	25.11	–
10^9	24.77	–
10^{10} (guardrail)	23.84	prior adopted base
10^{11}	22.62	–
10^{12}	21.20	–
10^{12} ETL side only	21.35	surveyed base
10^{12} HTL side only	21.20	surveyed base
10^{10}, $\sigma \times 10$ (verified)	22.21	matches $N \times 10$ (rate $\propto N\sigma$)
removed (noIF)	25.16	above detailed-balance Voc: unphysical

Table 13: Guardrailed regime in portable metrics ($\sigma = 10^{-19}$ m^2, $v_{th} = 10^5$ m/s): bulk lifetime τ and interface velocity S.

Density	τ / S	Note
$N_t = 10^{12}$ cm^{-3} (floor)	$\tau = 100$ μs	1–2 orders beyond best films
$N_t = 10^{13}$ cm^{-3}	$\tau = 10$ μs	–
$N_t = 10^{14}$ cm^{-3}	$\tau = 1$ μs	–
$N_t = 10^{15}$ cm^{-3} (routine)	$\tau = 0.1$ μs	routine case
$N_{if} = 10^{10}$ cm^{-2} (guardrail)	$S = 100$ cm/s	floor of stated box
$N_{if} = 10^{11}$ cm^{-2}	$S = 1000$ cm/s	–

Table 14: Guardrail audit: identical cells with interfaces ON versus OFF.

Cell	IF ON (%)	IF OFF (%)	Inflation (pts)
1.5 μm cell	23.40	24.87	+1.46
2.0 μm cell	23.47	24.73	+1.26
Champion (surveyed base)	23.50	25.16	+1.66

Table 15: Morris screen design and results ($r = 8$ trajectories, $p = 4$, $\Delta = 2/3$, seed 7; 56/56 converged). μ^* and σ are per normalised unit input; the per-decade column (sixth) and the CI column (seventh) are both per physical decade (per eV or per μm where noted); $n = 8$ unique directed effects per factor; CI = bootstrap 95% interval (5000 resamples), also per physical decade.

Factor	Range	μ^*	σ	n	per decade	95% CI
$\log N_{if}$ (both IF)	10^{8}–10^{12} cm^{-2}	4.08	1.08	8	1.02	[0.84, 1.22]
$\log N_t$ (absorber)	10^{12}–10^{15} cm^{-3}	2.05	1.40	8	0.68	[0.35, 0.99]
E_g (+χ link)	1.65–1.75 eV	1.75	0.79	8	17.5/eV	[12.6, 23.2]
Thickness	0.4–2.2 μm	1.29	0.99	8	0.72/μm	[0.36, 1.07]
$\log N_{D,ETL}$	10^{16}–10^{19} cm^{-3}	0.68	1.04	8	0.23	[0.02, 0.47]
$\log N_A$ (absorber)	10^{15}–3×10^{16} cm^{-3}	0.48	0.56	8	0.32	[0.12, 0.58]

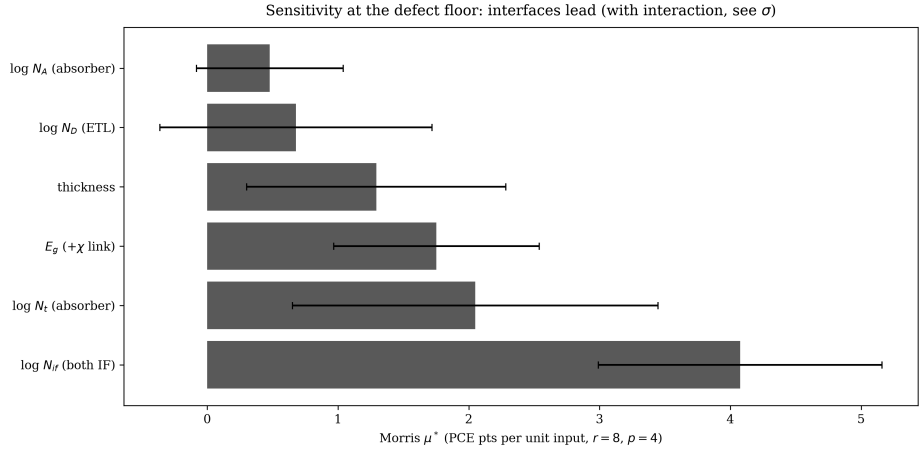


Figure 11: Morris μ^* (bars) with interaction spread σ (whiskers). Per-decade, per-eV, and per-μm figures are not commensurable across units; ranking is regime-conditional: interfaces lead at the defect floor, bulk defects at routine levels (Table 16); large σ signals interactions.

bandgap moves 17.5 points; per μm, thickness 0.72. These per-unit numbers are not commensurable across units, the interface figure bundles both heterojunctions over a 4-decade range against 3 for bulk, and—crucially—the E_g factor confounds gap with band offset, since the χ link assigns half of every gap step to the conduction band while Section 5.0 shows alignment alone moves efficiency by several points. The gap-step pattern betrays the confounding (all on the prior adopted base, 23.8365 anchor): 1.65 to 1.68 eV costs 1.13 points but 1.68 to 1.70 eV only 0.19 (and 1.70 to 1.72 eV costs 0.93), tracking the E_a landscape rather than the absorption edge. On the joint-optimum base the same steps even change sign (24.44 to 24.73% from 1.68 to 1.70 eV), confirming the steps map the activation landscape, not the gap. The random-forest/Saltelli corroboration ($S_T = 0.52$ interfaces, 0.33 bulk, 0.24 E_g; OOB $R^2 = 0.93$) is reported with the caveat that it is *not* independent evidence: the surrogate trains on all 190 standard-base runs including the Morris design itself, so its indices partly reflect the sampling design. The decisive test remains the dose–response repeated at routine defects: at $N_t = 10^{15}$ cm^{-3} and 1.0 μm, N_{if} 10^{9}–10^{11} cm^{-2} moves efficiency only from 20.84% to 20.67% (0.16 points over two decades), while the N_t 10^{14}–10^{15} step costs far more. At the defect floor interfaces dominate; at routine defects bulk defects do. Trap depth is a non-factor at this defect level: moving E_t to 0.3 or 0.9 eV changes efficiency by at most 0.012 points (23.8259% and 23.8241% against 23.8365%).

Interface-led sensitivity is consistent with passivation studies in which interface-adjacent defects gate the voltage [31], and with Bayesian optimisation on CsSnI3 where the optimal thickness likewise shifts with defect density [11].

Table 16: Routine-defect case ($N_t = 10^{15}$ cm^{-3}): interface dose–response is nearly flat, so bulk defects dominate here. Dose rows are 1.0 μm and Δ is against the 20.82% guardrail row of that base; the two ITO rows are 0.8 μm runs (20.72% at surveyed χ) and their Δ is against that 0.8 μm row, not the 1.0 μm one. Values computed; $R_s = 0$ unless stated.

N_{if} (cm^{-2})	PCE (%)	Δ vs guardrail	Base
10^9	20.84	+0.017	surveyed χ
10^{10} (guardrail)	20.82	–	surveyed χ
10^{11}	20.67	−0.147	surveyed χ
10^{10}, ITO 4.4 eV (0.8 μm)	20.68	−0.04	surveyed χ
10^{10}, ITO 4.4 eV (0.8 μm)	18.11	−2.61	adopted χ
10^{10}, $R_s = 1$ Ωcm^2	20.42	−0.40	surveyed χ; computed (s14)

9.0 Discussion

Fabrication plausibility bounds the result, and the computed sweeps price each term. $N_A = 3 \times 10^{16}$ cm^{-3} is routinely achievable; $E_g = 1.65$ eV sits below the ∼1.7 eV of the experimental records and contributes roughly one point (best 1.694 eV grid point: 22.506%); $N_t = 10^{12}$ cm^{-3} demands near-single-crystal quality ($\tau_{\mathrm{SRH}} \approx 100$ μs, Table 13). The contacts are load-bearing. At the surveyed anchor Ni 5.5 to 5.2 eV costs nothing (23.5041 to 23.5039%, the Ni 5.2 value reproduced digit-identically across two batches) while Ni 5.0 eV costs 0.25 points (23.2571%); on the joint-optimum affinities the same Ni 5.0 eV choice instead collapses the cell to 15.63% (Voc 1.01 V) against 18.83% at Ni 5.2 eV—a back-contact barrier cliff, not a slope, so the back-contact tolerance window collapses as the bands shift with affinity. ITO 4.0 to 4.7 eV collapses the cell to 18.60% (Voc 1.08 V)—SCAPS's thermionic-only contact boundary, so 4.0 eV is the edge of what this model tolerates rather than a typical ITO value; the routine-defect penalty is larger still, with ITO 4.4 eV costing 0.04 points at surveyed χ (20.72 to 20.68% on the 0.8 μm base) but 2.61 points at adopted χ (20.72 to 18.11% on the same base): the front-contact tolerance window itself shifts with ETL affinity (Table 16). Both optimum affinities are aggressive ($\chi_{\mathrm{TiO2}} = 3.7$ below the typical 4.0–4.2 eV window; $\chi_{\mathrm{CBTS}} = 3.9$ above the surveyed 3.6 eV) and move efficiency through the interface activation energy rather than transport (Table 6). At routine settings the model gives 20.82% (1.0 μm, $N_t = 10^{15}$ cm^{-3}, surveyed χ), 20.68% with ITO at 4.4 eV on the 0.8 μm base, 18.11% when that ITO is combined with the adopted χ_{TiO2}, and 20.42% with $R_s = 1$ Ωcm^2. Against experiment (Table 17): 24.95% exceeds the 22.02% certified record by 2.93 points; the excess over the best measured devices comes mostly from fill factor (∼89% simulated against ∼83% measured) and the low gap, not from voltage—and we do not attribute that FF excess to any single cause: the model carries no transport-layer bulk recombination and no back reflector, two systematic choices that both bias FF upward by amounts this study does not isolate, while the 0.017-point IV-step uncertainty only sets the significance threshold for any of these comparisons—our Voc deficit (0.33 V against E_g/q, 0.05 V against the radiative limit) is comparable to the 1.27–1.29 V record devices. With computed $R_s = 1$ Ωcm^2 (24.52%) the margin is 2.50 points, and at attainable gaps on the joint-optimum base the model gives 24.44% ($E_g = 1.68$ eV), 24.73% ($E_g = 1.70$ eV) and 24.28% ($E_g = 1.72$ eV), with 24.34% at $R_s = 1$ Ωcm^2 ($E_g = 1.70$ eV; $\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$, ITO 4.0 eV throughout). The non-monotonic 1.68-to-1.70 step (+0.29 points) is the E_a confounding of Section 8.0, not an absorption effect. The result sits at about four-fifths of the ∼30–31% Shockley–Queisser limit at 1.65 eV. A final calibration note: integrating the computed QE against the AM1.5G spectrum gives 22.39 mA/cm^2 against the 21.20 mA/cm^2 IV value—a 1.19 mA/cm^2 gap we report as Jsc uncertainty (headline Jsc uses the conservative IV value) alongside the ∼1 mA/cm^2 UV envelope.

The framework bites on live claims. A 2026 machine-learning-coupled screen reports 30.78% for a Sr_3PBr_3 absorber with a CBTS hole transport layer—our HTL family—at Voc 1.32 V, FF 87.05%, with SHAP importances led by acceptor and defect densities [32]. Three checklist rows fail on the

Table 17: Loss budget: joint optimum against measured record devices and model limits. FF is where the simulation exceeds experiment, not Voc. PCE values are each paper's headline; the Lei row is a mixed-source triple (its printed FF gives 22.24%, its reported headline 22.16%).

Device	Voc (V)	Jsc (mA/cm^2)	FF (%)	PCE (%)
This work (joint optimum)	1.321	21.20	89.11	24.95
AP-modified CsPbI3 [33]	1.29	20.77	83.01	22.16
QD-interface CsPbI3 [34]	1.28	–	–	22.15
Dual-interface CsPbI3 [5]	1.27	–	–	21.71
Carbon HTL-free [35] (cert. 20.35)	1.202	20.72	83.2	20.72
Radiative limit (this model)	1.371	39.2 (SQ)	90.5	–

published record: no interface status or dose is stated, the radiative coefficient is not reported, and the 30.78% headline sits above the ∼30% single-junction ceiling for its 1.8 eV gap class. Our audit suggests the direction of each correction (interfaces on, literature C_r, scripted parasitics), all pointing down—but those magnitudes are measured on CsPbI3 and are not transferable to Sr_3PBr_3, so no corrected number is quoted. We do not re-simulate their stack—different absorber, different gaps—so this is a reporting audit, not a recomputation; the point is that the checklist adjudicates a published 30%+ claim in one paragraph, which is what a standard is for.

10.0 Limitations and outlook

A laboratory translation would need, simultaneously: ∼1.0 μm CsPbI3 films at routine defect levels for the 20.86% case (time-resolved photoluminescence verified), interface passivation toward 10^{10} cm^{-2} following the fluorinated-dipole [4] or dual-interface [5] chemistries, a front contact near 4.0 eV, and $R_s \leq 1$ Ωcm^2 (Table 10). Not demonstrated here: 2.2 μm device-quality CsPbI3 at $N_t \sim 10^{12}$ cm^{-3}, and sputtered CBTS directly on CsPbI3 (CBTS typically requires high-temperature sulfurisation), for which we found no precedent—so this section is a requirements list, not a recipe. In-line controls, where applicable: QE edge at 752 nm (Figure 5) and the -0.16%/K temperature coefficient (Table 9).

Five boundaries delimit this work. First, $N_t = 10^{12}$ cm^{-3} at 2.2 μm demands near-single-crystal films, and with the modelled $\sigma = 10^{-19}$ m^2 implies $\tau_{\mathrm{SRH}} \approx 100$ μs, one to two orders beyond the best reported perovskite films; a reader preferring $\sigma = 10^{-17}$ m^2 would get 1 μs, so the lifetime claim scales as $1/\sigma$ and the defect density, not the lifetime, is what we hold fixed; at routine 10^{15} cm^{-3} a dedicated sweep peaks at 20.82% (1.0 μm), 20.68% with ITO at 4.4 eV on the 0.8 μm base, 18.11% when that ITO meets the adopted χ_{TiO2}, and 20.42% with $R_s = 1$ Ωcm^2. Second, the contacts (Ni 5.5 eV, ITO 4.0 eV), the joint-optimum affinities ($\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$ eV, both outside surveyed/typical ranges), and the barrier-side hole offset (+0.22 eV) are permissive-edge stated-box choices, each priced in Section 9.0. Third, phase stability of black CsPbI3 against the yellow δ-phase, ion migration, $E_g(T)$, and damp-heat degradation are outside SCAPS-1D steady-state scope and must be validated experimentally, following the passivation routes of [3–5]. Fourth, the grid is a screen plus a 104-configuration optimiser, not a proven global optimum: the best point sits at the corner of the allowed box, the exploratory search peaks 1.87 points below it among guardrail-satisfying runs, and the leading cluster is numerically inseparable. Fifth, the single-junction ceiling (∼30% at 1.65 eV [27]) caps how far any interface-inclusive screening can go.

11.0 Conclusion

Grid screening under interface-inclusive guardrails finds a 24.95%-efficient best point in the stated CsPbI3/CBTS box and 20.82% at routine defect levels, with every load-bearing assumption priced

Table 18: Guardrailed-reporting checklist: the minimum disclosure for a comparable SCAPS claim. The Protocol column maps each row to the standard simulation protocol of [1] (realistic inputs; full reporting; effective spectra and absorption; SQ check; explained deviations), whose optical half is implemented in [2].

Item	Disclose	This work	Protocol
Interfaces	status + N_{if} + σ (or S)	10^{10} cm^{-2}, 10^{-19} m^2 (S = 100 cm/s)	reporting
Bulk defects	N_t + σ (or τ)	10^{12} cm^{-3} (100 μs); routine 10^{15}	reporting
Radiative coeff.	C_r value used	0, priced at 10^{-10}/10^{-9} (Sec. 6.2)	inputs
Mobilities	μ_n/μ_p + anchor	25/25, priced at 10/5, 5/2 (Sec. 6.2)	inputs
Units	script vs file systems	Appendix A (two traps)	reporting
Parasitics	R_s/R_{sh} basis	scripted; routine case incl.	inputs
Contacts	both work functions	Ni 5.5 / ITO 4.0 eV, priced	inputs
Affinities	surveyed vs adopted χ	joint optimum, priced (χ-link)	no isolated opt.
Optics	R, absorption, spectrum	$R = 10\%$ flat; edge-moving Eg-sqrt; Jsc uncert. carried	spectra
SQ screen	Voc limit + deviations	1.371 V at 1.65 eV; no-IF disqualified	SQ check
Receipts	defs + raw outputs	public repo (no request step)	reporting

by computed sweep: single-crystal-grade defects, a low gap, a work-function-tuned front contact, joint affinity optimisation at aggressive values, and—new against the literature—the radiative-recombination default and the mobility values. Designed as the electrical-loss companion to the optical-loss framework of [2] and a worked implementation of the standard simulation protocol of [1], this study maps each known overestimation cause to one priced row. An exploratory search over a wider box found nothing higher, though by construction it does not bracket the champion. The same campaign prices the routine-defect thickness optimum, shows the interface dose–response to be regime-dependent (leading at the defect floor by a paired 2.03 points, nearly flat at routine defects), replaces an analytic resistance derate with scripted parasitic sweeps, checks the doping by Mott–Schottky self-consistency, and bounds the result by a model-derived radiative Voc that leaves the no-interface band without margin. Table 18 states the reporting standard as a concrete checklist. We recommend guardrailed reporting (interface status *with* dose level and defect level, defect floors *with* cross-sections, radiative coefficient *with* value, unit-system declaration, Rs basis, contact work functions, affinity choices) as a publication standard for SCAPS studies.

Data availability

Every artefact named in this paper is archived in the public repository `https://github.com/touhidsiddiqueeraj-bit/cspbi3-cbts-guardrailed`. Definition files (`defs/`) carry the base cell with interfaces, the no-interface variant, and the work-function and cross-section variants; `outputs/` holds every sweep receipt and raw curve, including the per-run logs; `scripts/` regenerates each receipt from the archived definitions; `receipts/` and `manuscript/` hold the verification receipts and the typeset source. Table 19 is the itemised inventory. Reference PDFs under `sources/` are copyrighted and listed in REFERENCES.md instead. This exceeds the "available on reasonable request" convention: there is no request step, because the definition files reproduce every number in this paper from a clean checkout plus SCAPS-1D 3.3.10 under WINE.

Author contributions

The authors designed the screening study, built the SCAPS definition files, executed the simulation campaign, performed the audit and sensitivity analyses, and wrote the manuscript.

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Appendix A: two scripting traps and a verification rule

Unit trap. SCAPS definition files store SI units (metres, m^{-3}, m^2/Vs) but script `set` commands expect practical units (μm, cm^{-3}, cm^2/Vs)—verified by dose–response: `set layer2.NA 1e15` reproduces the file baseline (stored 10^{21} m^{-3}) exactly, while SI-valued inputs (e.g. 10^{22}) inject a $10^6\times$ overdose and yield degenerate ghost cells (Voc 1.47–1.53 V, FF 91–92%, up to 27%). The same same-value check extends the map: `set layer2.radiative 0` and `set layer2.mun/mup 25/25` reproduce both headline receipts digit-identically, so radiative uses cm^3/s and mobilities cm^2/Vs in script units. Practical flags that catch them: Voc above the E_g-implied ceiling for the modelled gap, or FF above Green's ideal-diode value at the computed Voc.

Structural trap. `set interface<n>.IFdefect<k>.capture_cross_section` does not adjust σ in place; it appears to restructure the interface defect block, and in our hands it neutralised the addressed interfaces entirely (one interface: 23.8565%; both interfaces: 25.1494%, essentially the

no-interface value). The definition-file route (rewriting `sigma_nleft/.../pright`) reproduced the $N \times 10$ result to four decimals, which the set-key route could not.

Rule. For any parameter that changes a defect population, verify the applied value with a `get` of the same field in the same script and record the returned number in the run receipt; treat a result that moves efficiency in the wrong direction as a failed application, not a physical finding.

Appendix B: run inventory

All simulation batches are inventoried in Table 19.

Table 19: Simulation batches archived in the repository (run counts; solo reruns noted where they occurred).

Batch	Configurations	Receipt
Baseline validation	4 (one worker-crash retry)	`baseline_results.json`
Unit diagnostics	6 + 4 retries	`diag_units.json`
Round A grid	96 (3 worker-crash misses)	`roundA.jsonl`
Round B refine	36 (3 worker-crash misses)	`roundB.jsonl`
Characterisation + audit + T	17 + 6 re-runs	char.jsonl
Set-command verification	5	`s0_verify.json`
Interface dose + Rs/Rsh + WF	14	`s1_dose.json`
$N_t \times \sigma$ joint	3	`s4_sig.json` + `s4b_sig.json`
Morris screen	56 (0 errors)	`s5_morris.jsonl`
σ verify + χ_{CBTS} + Nt1e15 grid + realistic + noIF-JV	16 (1 worker retry)	`s6_rev2.json` + `noIF_JV.json`
Affinity adoption (χ_{TiO2}) + adopted-base anchors	12	`s7_r3.json` + `s8_champ38.json`
Morris bootstrap intervals	5000 resamples	`s5_morris_ci.json`
Routine-defect $\sigma \times 1$ baseline	1	`s9_nt15base.json`
Attainable gap + E_t sweep	6	`s10_r4.json`
Fine-IV confirmatory (surveyed base)	1	`s12_fine.json`
Joint affinities + ITO/Ni	52 (3 solo reruns merged)	`s11_joint_chi.json`
Box optimiser (LHS + pattern)	104 (9 worker-file misses)	`s13_opt.json`
Paired Morris bootstrap + τ/S	analysis only	`s13_analysis.json`
Routine Rs + adopted Rs/Rsh + QE	3	`s14.json` + `adopted_qe_raw.txt`
Affinity-peak extension + new-base T/Eg/Rs	26 + 6 (4 solo reruns merged)	`s16.json` + `s17.json`
Sobol-on-surrogate	RF, $N = 4096$ Saltelli	`sobol_surrogate.json`
Morris summary + CI + τ/S	derived, 5000 resamples	`s5_morris_summary.json` + `s5_morris_ci.json`
N_t 1e12 step retry	1	`retry_nt.json`
Analytic: DB Voc, QE integral, SQ Jsc	closed form	`s18_analytic.json`
Probes + set-key verification + analysis	4 + 5 + derived	`probe_qecv.json`, `s10_sigverify.json`, `analysis.json`
Thickness $\times$ N_t factorial	8 (2 solo reruns, deduped)	`s21_thNt.json`
Joint-optimum interface dose check	2	`s22_jointIF.json`
Radiative + mobility pricing (1 worker-crash retry)	11	`s23_krad_mu.json`
Raw per-run logs (jsonl)	–	`s13_opt.jsonl`, `s16.jsonl`